

BUSINESS INTELLIGENCE · AVIATION ANALYTICS

# High Cloud Airlines

## Performance Analytics Dashboard

*An Executive Business Intelligence Report presenting operational, commercial and network performance insights derived from U.S. airline operations data, delivered through an interactive Power BI analytics platform.*

PREPARED BY  
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ENGAGEMENT  
**Data Analytics  
Internship**

ORGANIZATION  
**AI Variant**

Power BI | Power Query | DAX | Data Modeling

**CONFIDENTIAL – PORTFOLIO EDITION**

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## SECTION 01

# Executive Summary

*This report presents the outcome of a Business Intelligence engagement undertaken to evaluate the operational and commercial performance of the U.S. domestic airline network between 2008 and 2013, using a Power BI analytics solution developed end-to-end – from raw data ingestion to executive-ready visualization.*

## Overview

The High Cloud Airlines Performance Analytics Dashboard was developed to convert large-scale, unstructured airline operations data into a structured, governed data model and a suite of interactive dashboards. The solution enables stakeholders to monitor passenger traffic, load factor, route and carrier performance, freight movement, and network reach through a single, unified analytical platform.

## Key Findings

- ◆ The network sustained an overall Load Factor of **76.80%**, reflecting efficient utilization of available seat capacity across the observed period.
- ◆ **187 million** passengers were transported across **2.80 million** flights, operated by **210 airlines** spanning approximately **18,000 routes**.
- ◆ Load Factor improved from **75.30% in 2008** to **78.12% in 2013**, indicating a consistent upward trend in operational efficiency.
- ◆ Weekday operations accounted for **71.41%** of total flights versus **28.59%** on weekends, highlighting a business-travel-driven demand pattern.
- ◆ Southwest Airlines led the network in flight volume, followed by Delta Air Lines and SkyWest Airlines.

### STRATEGIC TAKEAWAY

The analysis confirms a structurally healthy network with improving efficiency year-on-year. Opportunities exist to further optimize weekend capacity, strengthen underperforming routes, and extend the operating model of top-performing carriers across the broader network.

SECTION 02

# Business Problem

*Airlines generate vast volumes of operational data across ticketing, scheduling, and logistics systems. Without a consolidated analytical layer, this data remains fragmented and difficult to act upon.*

## The Challenge

Operational and commercial data at High Cloud Airlines was distributed across disconnected sources, making it difficult for decision-makers to answer fundamental performance questions in a timely manner:

- ◆ Which routes and carriers are performing above or below network average?
- ◆ How efficiently is available seat capacity being utilized over time?
- ◆ What seasonal and weekday/weekend patterns drive passenger demand?
- ◆ Which geographies and cities anchor the network's connectivity?
- ◆ How is freight volume distributed across logistics partners?

In the absence of a governed reporting layer, answering these questions required manual data extraction and consolidation — a process that is slow, error-prone, and unable to support timely, data-driven decisions.

## Impact of the Gap

|                                                                                                                                              |                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OPERATIONAL IMPACT</b><br>Delayed visibility into load factor and route efficiency limits the ability to reallocate capacity proactively. | <b>COMMERCIAL IMPACT</b><br>Without carrier and route benchmarking, high-performing patterns cannot be systematically identified or scaled. |
|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|

## The Need

A centralized, interactive Business Intelligence solution was required — one capable of transforming raw operational records into a governed data model, and presenting the resulting insights through a self-service dashboard suitable for executive and operational decision-making alike.

SECTION 03

# Business Objectives

The engagement was scoped around eight core analytical objectives, each translated into a corresponding dashboard capability.

| NO. | OBJECTIVE                        | DASHBOARD CAPABILITY                                                   |
|-----|----------------------------------|------------------------------------------------------------------------|
| 01  | Monitor passenger traffic trends | Time-series passenger volume analysis by month and year                |
| 02  | Track flight operations          | Flight volume trends, weekday vs. weekend split                        |
| 03  | Evaluate load factor             | Seat utilization efficiency across carriers, years, and distance bands |
| 04  | Assess route performance         | Top and bottom performing routes by volume and load factor             |
| 05  | Benchmark carrier performance    | Carrier-wise flight volume and efficiency comparison                   |
| 06  | Analyze freight transportation   | Freight volume ranking by logistics carrier                            |
| 07  | Map geographic reach             | Origin and destination city and country-level distribution             |
| 08  | Enable operational KPI tracking  | Consolidated KPI dashboard with interactive filtering                  |

GUIDING PRINCIPLE

Every dashboard element was built to answer a specific business question, ensuring the final solution functions as a decision-support tool rather than a static reporting artifact.

## SECTION 03 – CONTINUED

## Project Overview, As Configured

*The scope above is not an abstract wishlist – it is exactly what was configured as the opening page of the live Power BI report, giving every stakeholder the same starting context.*

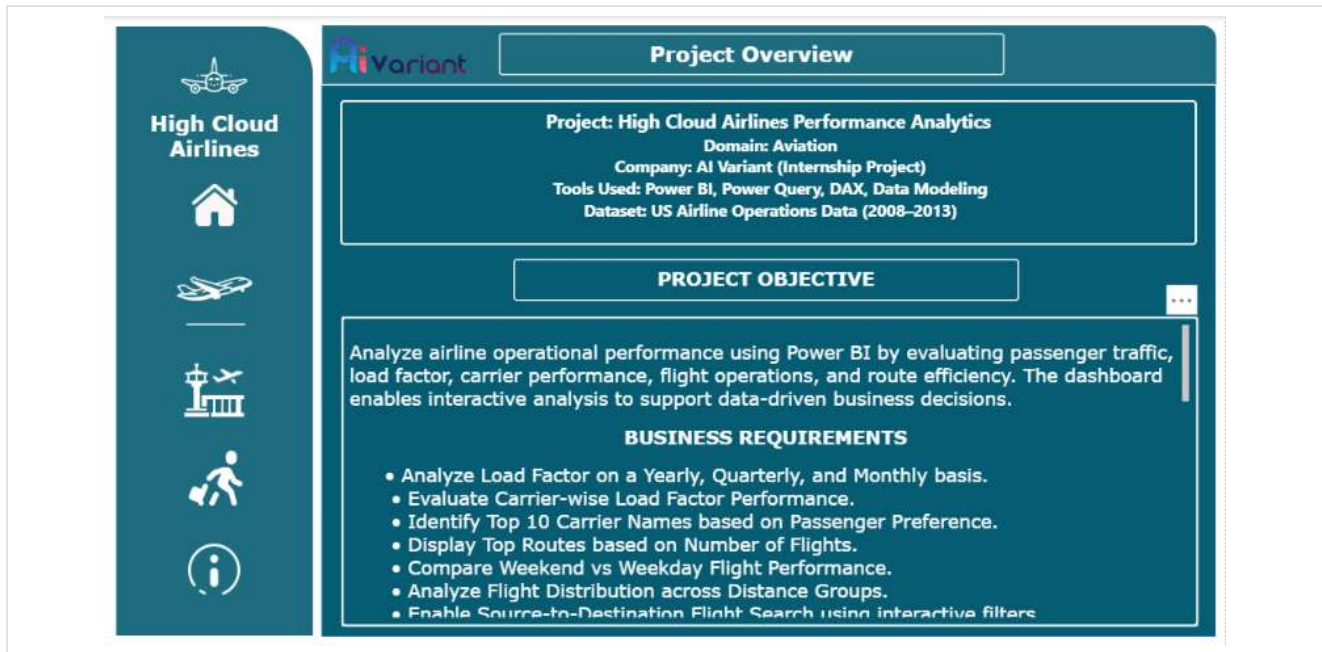


Fig. 1 – Project Overview Page, High Cloud Airlines Performance Analytics Dashboard

### TOOLS CONFIGURED

Power BI · Power Query · DAX · Data Modeling — applied against the 2008–2013 U.S. Airline Operations dataset, exactly as declared on the report's own overview page.

SECTION 04

# Dataset Overview

The analysis is built on a multi-year U.S. domestic airline operations dataset, covering flight-level, carrier-level, and route-level records between 2008 and 2013.

| Dataset Characteristics |                                       | Core Data Fields                                                                                                                                                                                                                                                                                                                    |
|-------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ATTRIBUTE               | DETAIL                                |                                                                                                                                                                                                                                                                                                                                     |
| Time Period             | 2008 – 2013                           |                                                                                                                                                                                                                                                                                                                                     |
| Geography               | United States (cross-border carriers) |                                                                                                                                                                                                                                                                                                                                     |
| Granularity             | Flight / Route / Carrier level        |                                                                                                                                                                                                                                                                                                                                     |
| Origin Cities           | 1,296                                 |                                                                                                                                                                                                                                                                                                                                     |
| Destination Cities      | 1,327                                 |                                                                                                                                                                                                                                                                                                                                     |
| Total Airlines          | 210                                   |                                                                                                                                                                                                                                                                                                                                     |
| Total Routes            | ~18,000                               |                                                                                                                                                                                                                                                                                                                                     |
|                         |                                       | <ul style="list-style-type: none"><li>◆ Flight date, carrier code and carrier name</li><li>◆ Origin and destination city / airport</li><li>◆ Scheduled distance and distance group</li><li>◆ Available seats and passengers carried</li><li>◆ Freight volume transported</li><li>◆ Day-of-week and seasonality indicators</li></ul> |

## Data Quality & Preparation

Prior to modeling, the raw dataset was profiled and cleansed using Power Query to standardize carrier and city naming conventions, resolve missing and duplicate records, correct data types, and engineer derived fields such as Load Factor, Distance Group, and Weekday/Weekend flags required for downstream analysis.

|                                                                                                                                                                    |                                                                                                                                                                             |                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>01</div> <div>Profile</div> <div>Audited every field for nulls, duplicates, inconsistent naming, and incorrect data types across the raw source tables.</div> | <div>02</div> <div>Cleanse &amp; Standardize</div> <div>Resolved carrier and city naming inconsistencies, corrected data types, and removed duplicate flight records.</div> | <div>03</div> <div>Engineer</div> <div>Derived Load Factor, Distance Group, and Weekday/Weekend classification fields required for downstream analysis.</div> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

### WHY THIS MATTERS

A rigorously cleansed dataset is the foundation of every downstream KPI in this report — without standardized carrier names and validated distance/seat figures, Load Factor and route-level comparisons would silently misstate performance.

SECTION 05

# Data Modeling Overview

A star-schema data model was implemented in Power BI to ensure query performance, calculation accuracy, and long-term maintainability of the analytical solution.

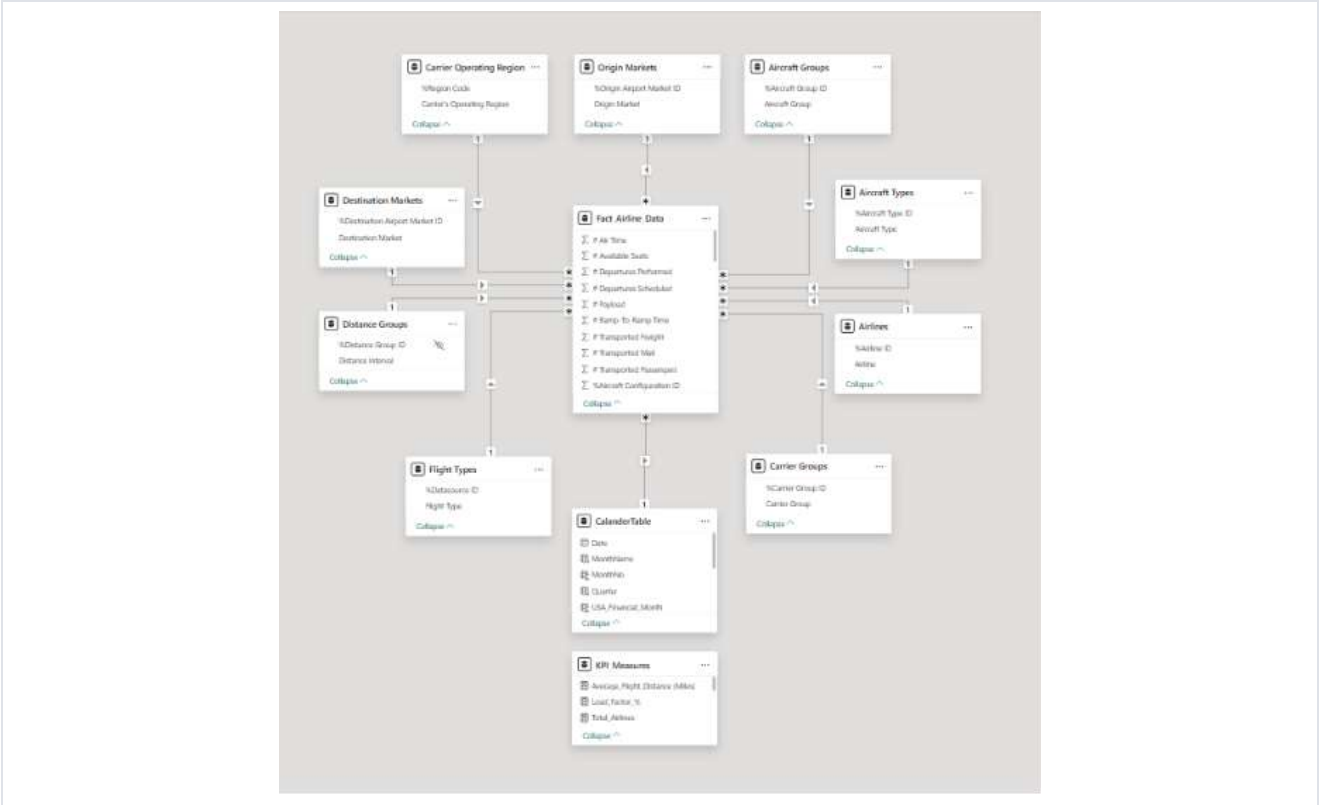


Fig. 2 – Power BI Model View, High Cloud Airlines Performance Analytics

## Modeling Approach

The model centers on a single fact table, **Fact Airline Data**, holding flight-level measures – Air Time, Available Seats, Departures Performed, Departures Scheduled, Payload, Ramp-to-Ramp Time, Transported Freight, Transported Mail, and Transported Passengers – connected to eight supporting dimension tables:

|                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>◆ CalendarTable – date intelligence and weekday/weekend classification</li><li>◆ Airlines &amp; Carrier Groups – carrier identity and grouping</li><li>◆ Origin Markets &amp; Destination Markets – route geography</li></ul> | <ul style="list-style-type: none"><li>◆ Distance Groups – mileage-band segmentation</li><li>◆ Aircraft Types &amp; Aircraft Groups – fleet classification</li><li>◆ Flight Types &amp; Carrier Operating Regions – operational segmentation</li></ul> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## SECTION 05 – CONTINUED

# Data Model, DAX & Date Intelligence

All relationships are single-direction, one-to-many links from each dimension into the fact table, following Power BI star-schema best practice to optimize DAX performance and avoid ambiguous filter paths. A dedicated KPI Measures table houses all calculated measures, keeping business logic separate from raw data.

## CalendarTable – Date Intelligence Fields

| Date       | Year | MonthName | Quarter | YearMonth | WeekdayName | USA_Financial_Month | USA_Financial_Quarter | Weekend_vs_Weekday |
|------------|------|-----------|---------|-----------|-------------|---------------------|-----------------------|--------------------|
| 01-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Tuesday     | 01-Q4               | WeekDay               |                    |
| 02-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Wednesday   | 01-Q4               | WeekDay               |                    |
| 03-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Thursday    | 01-Q4               | WeekDay               |                    |
| 04-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Friday      | 01-Q4               | WeekDay               |                    |
| 05-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Saturday    | 01-Q4               | WeekEnd               |                    |
| 06-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Sunday      | 01-Q4               | WeekEnd               |                    |
| 07-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Monday      | 01-Q4               | WeekDay               |                    |
| 08-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Tuesday     | 01-Q4               | WeekDay               |                    |
| 09-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Wednesday   | 01-Q4               | WeekDay               |                    |
| 10-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Thursday    | 01-Q4               | WeekDay               |                    |
| 11-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Friday      | 01-Q4               | WeekDay               |                    |
| 12-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Saturday    | 01-Q4               | WeekEnd               |                    |
| 13-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Sunday      | 01-Q4               | WeekEnd               |                    |
| 14-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Monday      | 01-Q4               | WeekDay               |                    |
| 15-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Tuesday     | 01-Q4               | WeekDay               |                    |
| 16-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Wednesday   | 01-Q4               | WeekDay               |                    |
| 17-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Thursday    | 01-Q4               | WeekDay               |                    |
| 18-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Friday      | 01-Q4               | WeekDay               |                    |
| 19-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Saturday    | 01-Q4               | WeekEnd               |                    |
| 20-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Sunday      | 01-Q4               | WeekEnd               |                    |
| 21-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Monday      | 01-Q4               | WeekDay               |                    |
| 22-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Tuesday     | 01-Q4               | WeekDay               |                    |
| 23-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Wednesday   | 01-Q4               | WeekDay               |                    |
| 24-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Thursday    | 01-Q4               | WeekDay               |                    |
| 25-01-2000 | 2000 | January   | Q1      | 2000-Jan  | Friday      | 01-Q4               | WeekDay               |                    |

Fig. 3 – CalendarTable, with MonthName, Quarter, USA\_Financial\_Month, WeekDayName, and Weekend\_vs\_Weekday fields

## KPI\_Measures – Core DAX Library

| Measure                         |
|---------------------------------|
| Average_Flight_Distance (Miles) |
| Load_Factor_%                   |
| Total_Airlines                  |
| Total_Available_Seats           |
| Total_Destination_Cities        |
| Total_Routes                    |
| Total_Transported_Freight       |
| Total_Origin_Cities             |
| Total_Flights                   |

Fig. 4 – KPI\_Measures table: 9 governed measures powering every visual in the dashboard

SECTION 06

# Power BI Development Process

The solution was developed following a structured, five-stage Business Intelligence methodology, from raw data acquisition through to executive dashboard delivery.

## 01 Data Acquisition

U.S. airline operations dataset (2008–2013) sourced and loaded into Power BI via Power Query.

## 02 Data Cleaning & Transformation

Standardized fields, handled nulls and duplicates, engineered Load Factor, Distance Group, and Weekday/Weekend attributes.

## 03 Data Modeling

Built a star-schema model with fact and dimension tables; defined relationships and hierarchies.

## 04 DAX Measure Development

Authored calculated measures for KPIs, time intelligence, and dynamic ranking logic.

## 05 Dashboard Design & Deployment

Designed a multi-page interactive dashboard with slicers, navigation, and cross-filtering for business consumption.

### Tooling by Stage

| STAGE                     | PRIMARY TOOL         | OUTPUT                                                |
|---------------------------|----------------------|-------------------------------------------------------|
| Data Acquisition          | Power Query          | Raw tables loaded into the Power BI data model        |
| Cleaning & Transformation | Power Query (M)      | Standardized, deduplicated, type-corrected tables     |
| Data Modeling             | Power BI Model View  | Star-schema with 8 dimension tables + fact table      |
| DAX Measure Development   | DAX                  | A governed KPI_Measures table of 9 core measures      |
| Dashboard Design          | Power BI Report View | 6-page interactive report with navigation and slicers |

#### METHODOLOGY NOTE

Each stage was validated before moving to the next — measures were spot-checked against raw aggregates, and visuals were cross-filtered end-to-end to confirm relationship

integrity across all eight dimension tables before the dashboard was considered release-ready.

SECTION 07

# Dashboard Overview

The final solution is a six-page, interactive Power BI dashboard designed for both executive summary consumption and analyst-level drill-down.

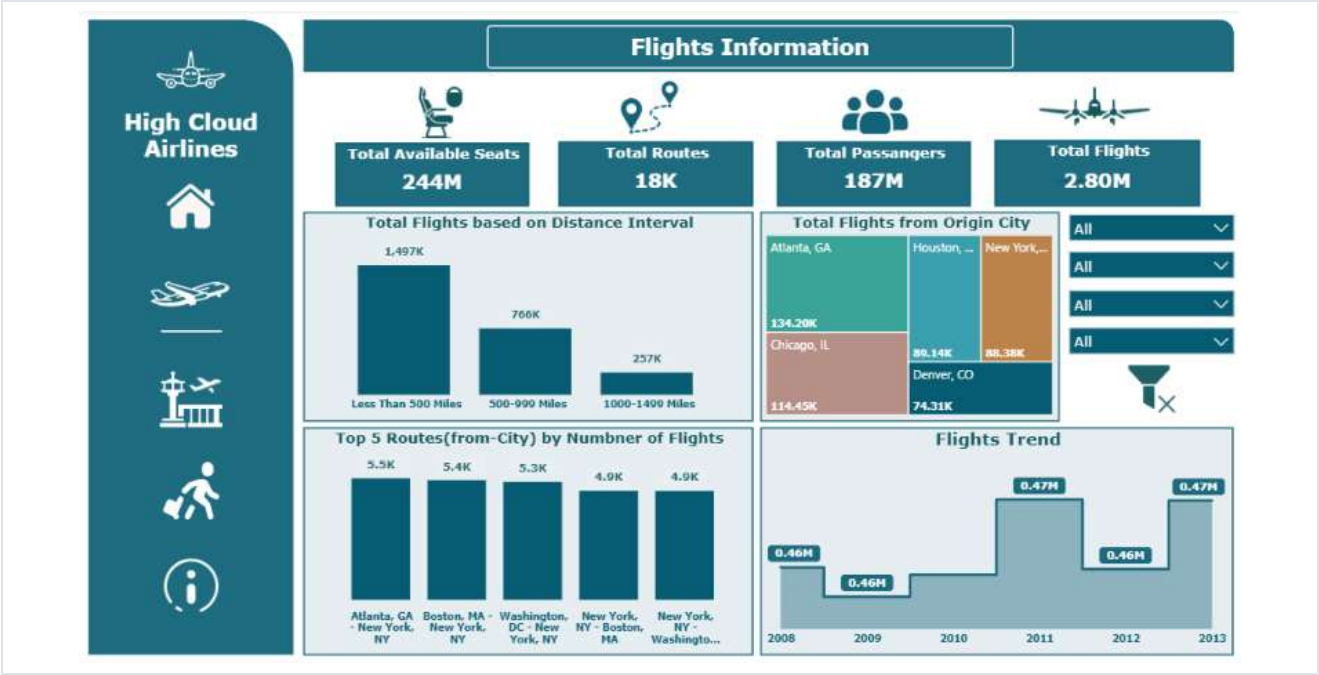


Fig. 5 – Flight Information Page: Seats, Routes, Passengers, Flights, Top Routes & Trend

## Design Principles

- |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>◆ Consistent navigation sidebar across all six report pages</li><li>◆ KPI cards positioned for immediate executive scanning</li><li>◆ Cross-filtering enabled across all visuals</li></ul> | <ul style="list-style-type: none"><li>◆ Minimal, corporate teal-and-white color palette</li><li>◆ Interactive slicers for year, carrier, country and weekday</li><li>◆ A dedicated Insights page translating KPIs into narrative</li></ul> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

SECTION 08

# KPI Summary

The dashboard surfaces ten headline KPIs, each interpreted below in business terms.

|                                                  |                                                   |                                             |                                                |
|--------------------------------------------------|---------------------------------------------------|---------------------------------------------|------------------------------------------------|
| <div>76.80%</div> <div>OVERALL LOAD FACTOR</div> | <div>244M</div> <div>TOTAL AVAILABLE SEATS</div>  | <div>187M</div> <div>TOTAL PASSENGERS</div> | <div>2.80M</div> <div>TOTAL FLIGHTS</div>      |
| <div>210</div> <div>TOTAL AIRLINES</div>         | <div>18K</div> <div>TOTAL ROUTES</div>            | <div>1,296</div> <div>ORIGIN CITIES</div>   | <div>1,327</div> <div>DESTINATION CITIES</div> |
| <div>8.49B</div> <div>FREIGHT TRANSPORTED</div>  | <div>743 mi</div> <div>AVG. FLIGHT DISTANCE</div> |                                             |                                                |

## Business Interpretation

| KPI                    | WHAT IT TELLS THE BUSINESS                                                                 |
|------------------------|--------------------------------------------------------------------------------------------|
| Load Factor (76.80%)   | Seat inventory is being utilized efficiently; a healthy benchmark for a network this size. |
| Total Flights (2.80M)  | Reflects the scale of operations across the analyzed period, informing capacity planning.  |
| Total Routes (18K)     | Indicates broad network reach and route diversification opportunity.                       |
| Freight (8.49B)        | Highlights the materiality of cargo operations alongside passenger revenue.                |
| Avg. Distance (743 mi) | Suggests a network weighted toward short-to-medium haul domestic routes.                   |

## SECTION 09

# Dashboard Pages Explained

The dashboard is organized into six focused analytical pages, each addressing a distinct business question.

## 1 Carrier Performance & Weekday Analysis

Ranks carriers by flight volume — led by Southwest Airlines, Delta Air Lines, and SkyWest Airlines — and compares weekday (76.73%) versus weekend (76.96%) Load Factor, alongside distance-band efficiency and the 2008–2013 trend line.

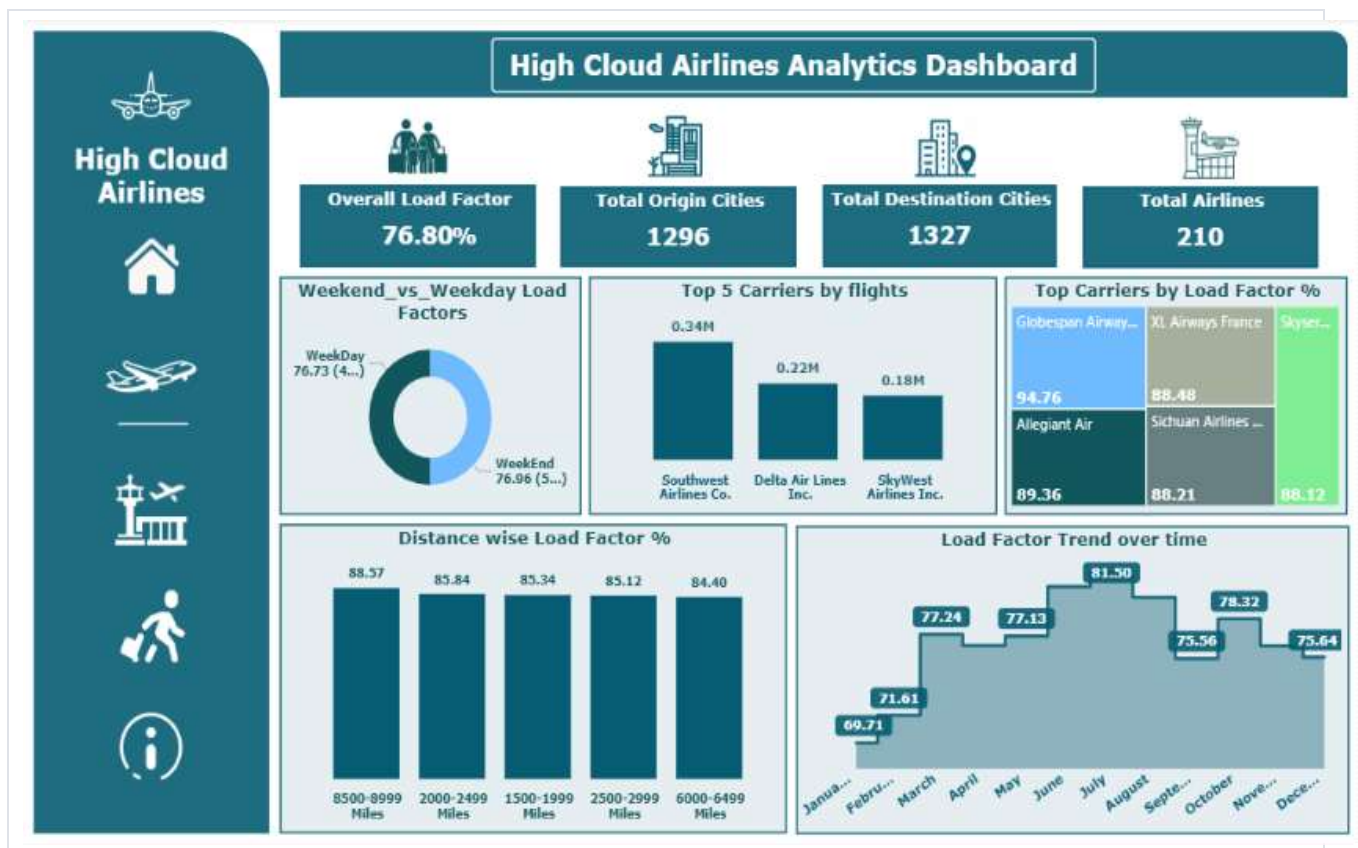


Fig. 6 – Home Page: Carrier Ranking, Weekday vs. Weekend Load Factor

2
Load Factor Analysis

Breaks down seat utilization by carrier and distance interval, with a year-over-year trend rising from 75.30% (2008) to 78.12% (2013). Globespan Airways leads all carriers at 94.76% Load Factor.

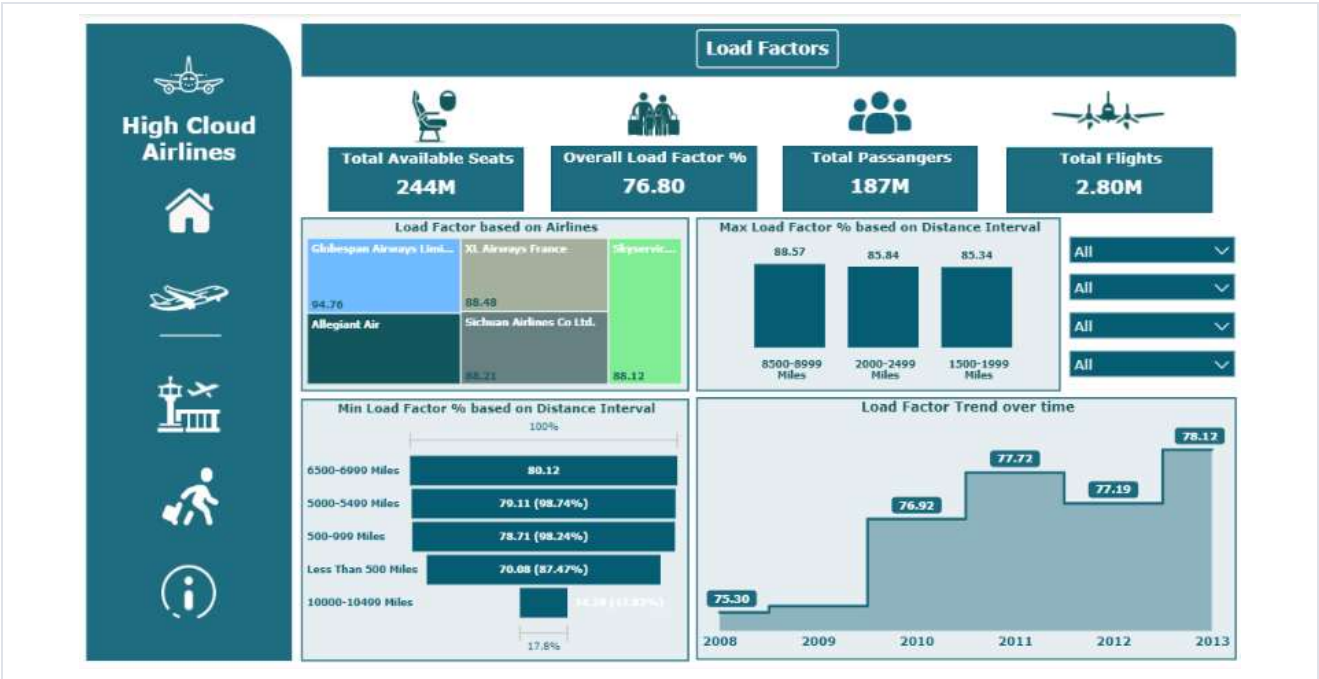


Fig. 7 – Load Factors Page: Carrier & Distance-Interval Breakdown

3 Passengers & Flights

Tracks Total Flights, Passengers, Freight, and Average Flight Distance alongside a multi-year Passengers trend line, the Weekend vs. Weekday flight split (28.59% / 71.41%), and airline distribution by country.

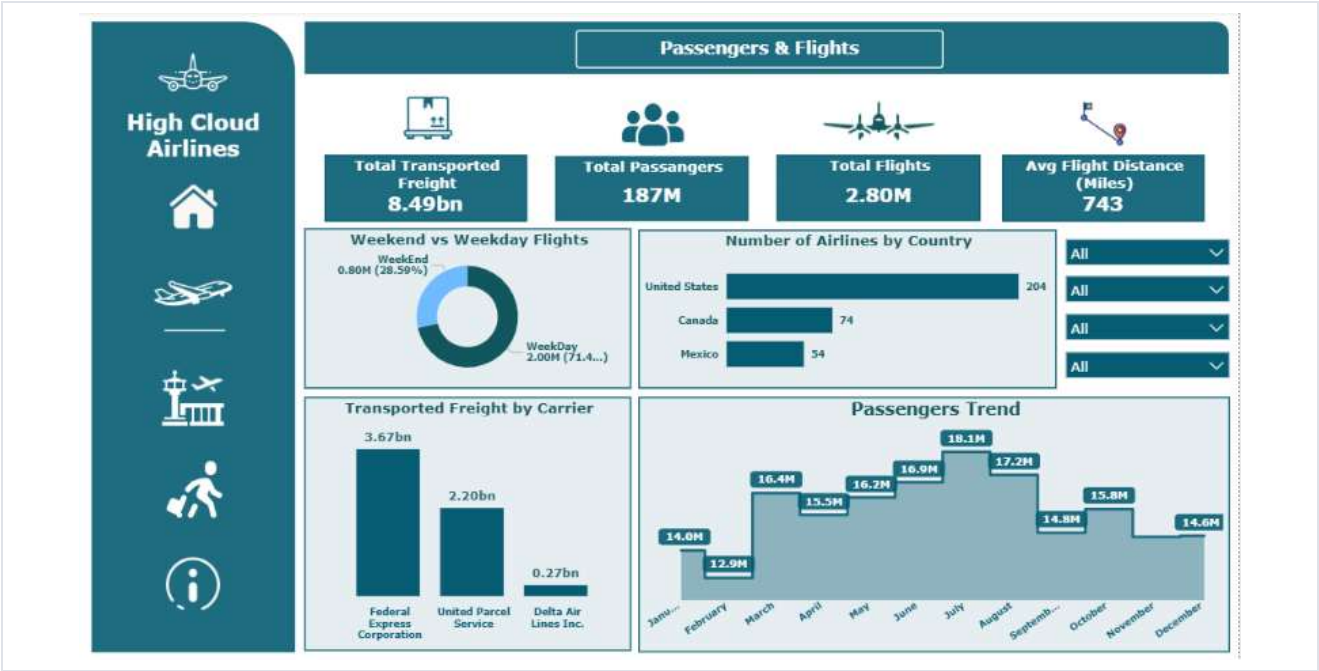


Fig. 8 – Passengers & Flights Page: Trend Analysis and Operational Split

Interactivity Summary

| FEATURE                                    | BUSINESS PURPOSE                                                |
|--------------------------------------------|-----------------------------------------------------------------|
| Year / Carrier / Country / Weekday Slicers | Enables filtered, role-specific analysis                        |
| Cross-Filtering                            | Selecting one visual dynamically updates all others on the page |
| Navigation Sidebar                         | Guided movement between report pages for non-technical users    |

4 Insights Summary

A dedicated narrative page translating the dashboard's operational KPIs into plain-language business insights – the same summary layer an executive would read before opening the interactive report.

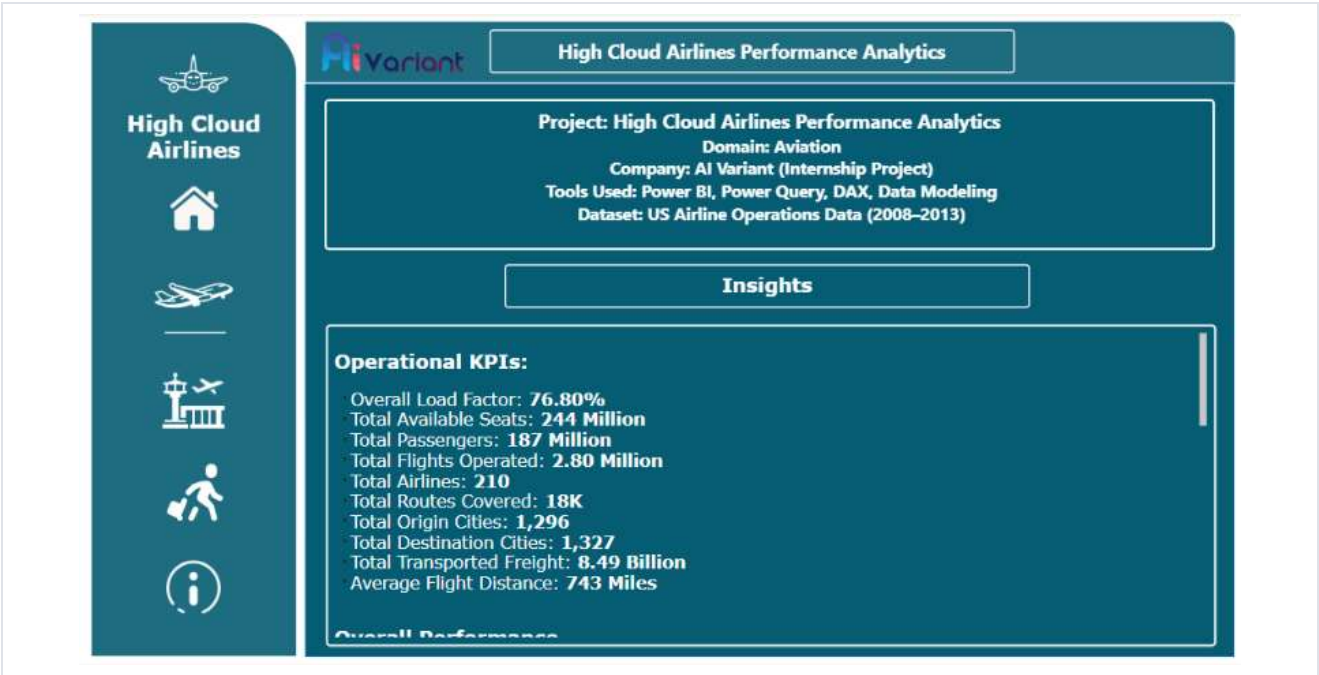


Fig. 9 – Insights Page: Operational KPI Narrative

## SECTION 10

## Business Insights

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*Six categories of insight emerged from the analysis, each with direct implications for operational and commercial strategy.*

### Load Factor Insights

- ◆ Load Factor improved steadily from **75.30% (2008) to 78.12% (2013)**, a positive efficiency trend.
- ◆ **Globespan Airways** recorded the network's highest Load Factor at **94.76%**.
- ◆ The 8,500–8,999 mile distance band showed the highest Load Factor among displayed distance groups, suggesting strong demand on ultra-long-haul segments.

### Passenger Insights

- ◆ Passenger traffic peaked in July, consistent with summer travel seasonality.
- ◆ February recorded the lowest passenger volume of the year.
- ◆ Demand follows a clear, repeatable seasonal curve suitable for predictive capacity planning.

### Flight Operations Insights

- ◆ Weekday flights represented **71.41%** of total volume versus **28.59%** on weekends.
- ◆ The weekday skew signals a network oriented toward business and routine travel.

## Carrier Insights

- ◆ Southwest Airlines operated the highest number of flights in the network.
- ◆ Delta Air Lines ranked second by flight volume.
- ◆ SkyWest Airlines featured among the network's leading carriers by operational scale.

## Geographical Insights

- ◆ The United States hosted the highest concentration of operating airlines.
- ◆ Canada and Mexico followed as the next most represented countries in the network.

## Freight Insights

- ◆ Federal Express transported the highest freight volume across the network.
- ◆ United Parcel Service ranked second by cargo volume.

### SYNTHESIS

Efficiency gains, seasonal predictability, and carrier concentration together suggest a network positioned to grow selectively — by extending high-Load-Factor practices to underperforming routes and carriers.

SECTION 11

# Recommendations

Based on the analysis, seven strategic actions are recommended to sustain and extend network performance.

| NO. | RECOMMENDATION                                | RATIONALE                                                                              |
|-----|-----------------------------------------------|----------------------------------------------------------------------------------------|
| 01  | Increase flights on high-demand routes        | Capture unmet demand identified through route-level passenger volume analysis          |
| 02  | Improve performance on low-utilization routes | Address routes with Load Factor materially below network average                       |
| 03  | Benchmark top-performing carriers             | Replicate operating practices from carriers such as Globespan Airways                  |
| 04  | Optimize weekend operations                   | Weekend flights are underrepresented at 28.59% of volume, indicating capacity headroom |
| 05  | Expand high-performing routes                 | Scale capacity on routes already demonstrating strong Load Factor                      |
| 06  | Improve freight planning                      | Align cargo capacity with demand patterns from leading freight partners                |
| 07  | Monitor Load Factor continuously              | Sustain the 2008–2013 efficiency improvement trend through ongoing tracking            |

SECTION 12

# Business Value

*The dashboard delivers measurable value across operational, commercial, and strategic dimensions of the business.*

**OPERATIONAL**

Monitor airline performance and track KPIs in real time, improving seat utilization and route optimization.

**COMMERCIAL**

Identify high-performing carriers and analyze passenger demand to inform revenue strategy.

**LOGISTICS**

Monitor freight operations and align cargo capacity with demand across partner carriers.

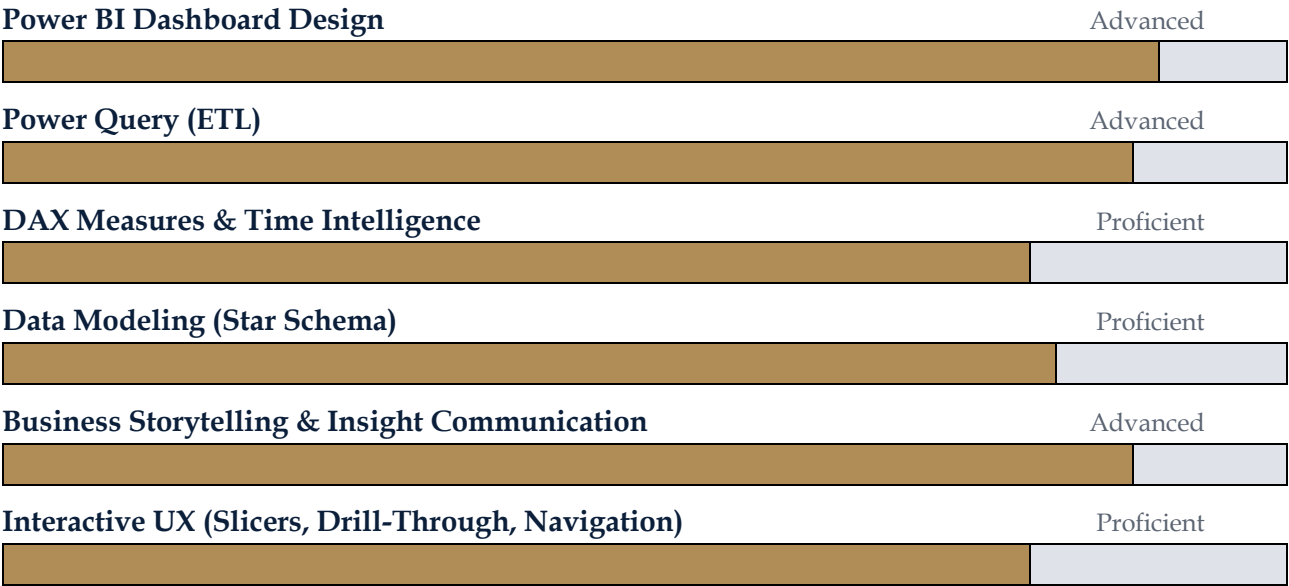
**STRATEGIC**

Support strategic planning, improve operational efficiency, and enable faster, data-driven decision-making.

SECTION 13

# Skills Demonstrated

*This engagement demonstrated end-to-end Business Intelligence capability, from data engineering through to executive storytelling.*



## Toolset

Power BI · Power Query · DAX · Data Modeling · Star Schema Design · Business Analysis · Data Storytelling

### WHY IT MATTERS TO AN EMPLOYER

This project was not built as an exercise — it mirrors the exact workflow a Data Analyst or MIS professional runs in production: raw data in, governed model out, decision-ready dashboard delivered on a deadline.

### APPLIED, NOT THEORETICAL

Every measure in the KPI\_Measures library was tested against manual aggregates before being trusted in a visual — the same discipline applied to MIS reporting in a live dealership environment.

SECTION 14

# Project Outcome

*The engagement delivered a fully functional, interactive Business Intelligence solution meeting all defined business objectives.*

|                                          |                                        |                                         |                                                 |
|------------------------------------------|----------------------------------------|-----------------------------------------|-------------------------------------------------|
| <b>6</b><br>DASHBOARD PAGES<br>DELIVERED | <b>9</b><br>CORE DAX MEASURES<br>BUILT | <b>8</b><br>DIMENSION TABLES<br>MODELED | <b>11</b><br>BUSINESS CAPABILITIES<br>DELIVERED |
|------------------------------------------|----------------------------------------|-----------------------------------------|-------------------------------------------------|

## Capabilities Delivered

Date Intelligence · Load Factor Analysis · Carrier Performance · Top Routes · Passenger Analysis · Flight Trends · Weekday/Weekend Split · Distance Groups · Geographic Analysis · Interactive Filtering · KPI Dashboard

## Deliverables

|                                                                                                                                                                         |                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DATA MODEL</b><br>A governed, star-schema Power BI data model built on the 2008–2013 airline operations dataset, with 8 dimension tables and a validated fact table. | <b>INTERACTIVE DASHBOARD</b><br>A 6-page interactive report — Project Overview, Home, Flight Information, Passengers & Flights, Load Factors, and Insights — with full cross-filtering. |
| <b>DAX MEASURE LIBRARY</b><br>A documented KPI_Measures table supporting time intelligence, load factor calculation, and dynamic ranking.                               | <b>EXECUTIVE REPORT</b><br>This document — summarizing methodology, insights, and recommendations for stakeholder and portfolio use.                                                    |

SECTION 15

# Conclusion

*The High Cloud Airlines Performance Analytics Dashboard demonstrates how a well-structured Business Intelligence solution can convert large-scale operational data into clear, actionable insight.*

Across the 2008–2013 period, the network sustained an improving Load Factor, a seasonally predictable demand pattern, and a small group of carriers driving disproportionate operational scale and efficiency. These findings, and the recommendations built upon them, illustrate the value of a governed analytics layer in supporting airline decision-making — from route planning to carrier benchmarking to freight strategy.

Beyond its analytical findings, this project reflects a complete, professional Business Intelligence workflow: data acquisition, cleansing, modeling, DAX-based calculation, interactive dashboard design, and executive-level insight communication.

*“The strongest dashboards don't just display numbers — they answer the question a decision-maker was already asking.”*

## Key Takeaways at a Glance

|                                                                                                                            |                                                                                                     |                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>76.80%</b><br>Network Load Factor, up from 75.30% in 2008 — a sustained efficiency gain worth protecting and extending. | <b>71:29</b><br>Weekday-to-weekend flight ratio, revealing clear headroom to grow weekend capacity. | <b>Top 3</b><br>Carriers — Southwest, Delta, SkyWest — anchor network scale and set the benchmark for efficiency. |
|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|

THANK YOU

# High Cloud Airlines Performance Analytics Dashboard

Prepared by Shubham Vishwakarma · Data Analytics Internship · AI Variant

Power BI | Power Query | DAX | Data Modeling